FORM TP 2014056

TEST CODE **01212032**

MAY/JUNE 2014

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE® EXAMINATION

CHEMISTRY

Paper 032 – General Proficiency

Alternative to SBA

2 hours and 10 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of THREE questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. DO NOT write in the margins.
4. Where appropriate, ALL WORKING MUST BE SHOWN in this booklet.
5. You may use a silent, non-programmable calculator to answer questions.
6. If you need to re-write any answer and there is not enough space to do so on the original page, you must request extra lined pages from the invigilator. **Remember to draw a line through your original answer and correctly number your new answer in the box provided.**
7. **If you use extra pages you MUST write your registration number and question number clearly in the boxes provided at the top of EVERY extra page.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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Answer ALL questions.

Write your responses in the spaces provided in this booklet.

1. Energy changes always accompany chemical reactions. When the energy change is in the form of heat, reactions can be investigated by using temperature measurements. John and Mary were asked to determine a value for the enthalpy change that accompanies a neutralization reaction. They were provided with chemicals and apparatus as listed below:

- 1.0 mol dm^{-3} sodium hydroxide solution
- 1.0 mol dm^{-3} sulfuric acid solution
- One 0–100 °C thermometer
- One 250-cm^3 Styrofoam cup and a Styrofoam lid (with a small hole for the thermometer)
- One 100-cm^3 measuring cylinder
- One stop-watch

The following procedure was followed:

(i) Allow the solutions of acid and base to equilibrate at room temperature.

(ii) Measure 100 cm^3 of the sodium hydroxide solution, pour it into the Styrofoam cup, cover with the lid, and record the temperature for **four** one-minute intervals.

(iii) On the **fifth** minute, add 50 cm^3 of the sulfuric acid solution to the Styrofoam cup.

(iv) Put on the Styrofoam lid, insert the thermometer and stir the solution.

(v) Record the temperature at 5.5 minutes, and at every 0.5–minute interval up to 10 minutes.

Some of the temperature readings are shown in Table 1.

TABLE 1: TIME-TEMPERATURE DATA

Time (min)	Temperature (°C)
1.0	27.0
2.0	27.1
3.0	27.1
4.0	27.1
5.0	
5.5	
6.0	
6.5	

Time (min)	Temperature (°C)
7.0	35.9
7.5	36.4
8.0	
8.5	36.7
9.0	
9.5	36.8
10.0	36.8

(a) From the diagrams in Figure 1, read the temperature at the times shown and insert the readings in Table 1.

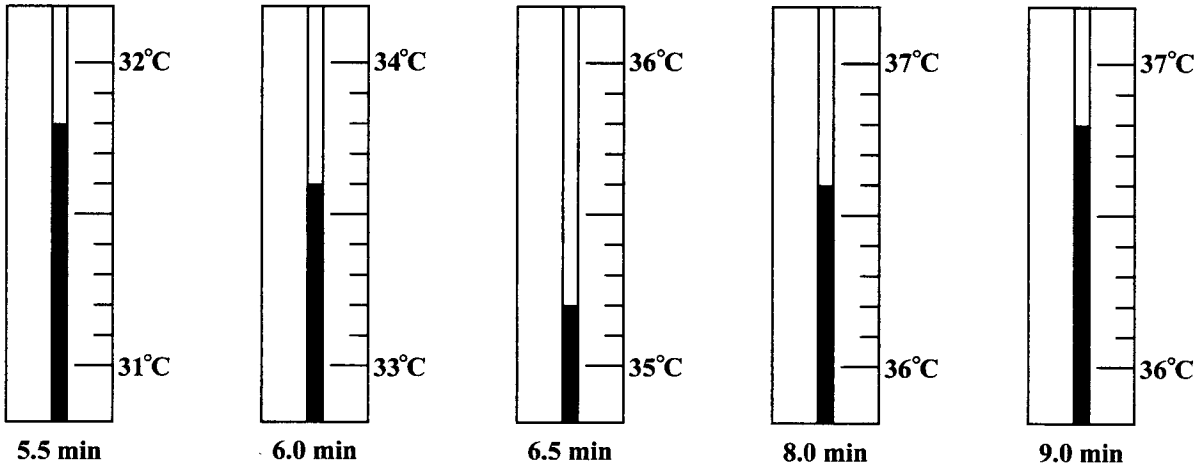

Figure 1. Thermometer readings at different times for the reaction of 1.0 mol dm^{-3} sulfuric acid with 1.0 mol dm^{-3} sodium hydroxide

(5 marks)

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(b) (i) Using the axes provided in Figure 2 on page 5, plot a graph of temperature against time using the data in Table 1. Some of the points have been plotted for you.

(3 marks)

(ii) Draw a smooth curve through the points on the graph in (b) (i). **(1 mark)**

(iii) From the graph, calculate the change in temperature for the reaction taking place in the Styrofoam cup.

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(1 mark)

(iv) State whether the reaction is exothermic or endothermic.

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(1 mark)

(c) (i) Write a balanced equation for the reaction in the Styrofoam cup.

Equation: .....

(2 marks)

(ii) Calculate the number of moles of sulfuric acid in the 50 cm^3 of 1.0 mol dm^{-3} H_2SO_4 that were used in the reaction.

(2 marks)

(iii) Determine the number of moles of water that were produced in the reaction.

(1 mark)

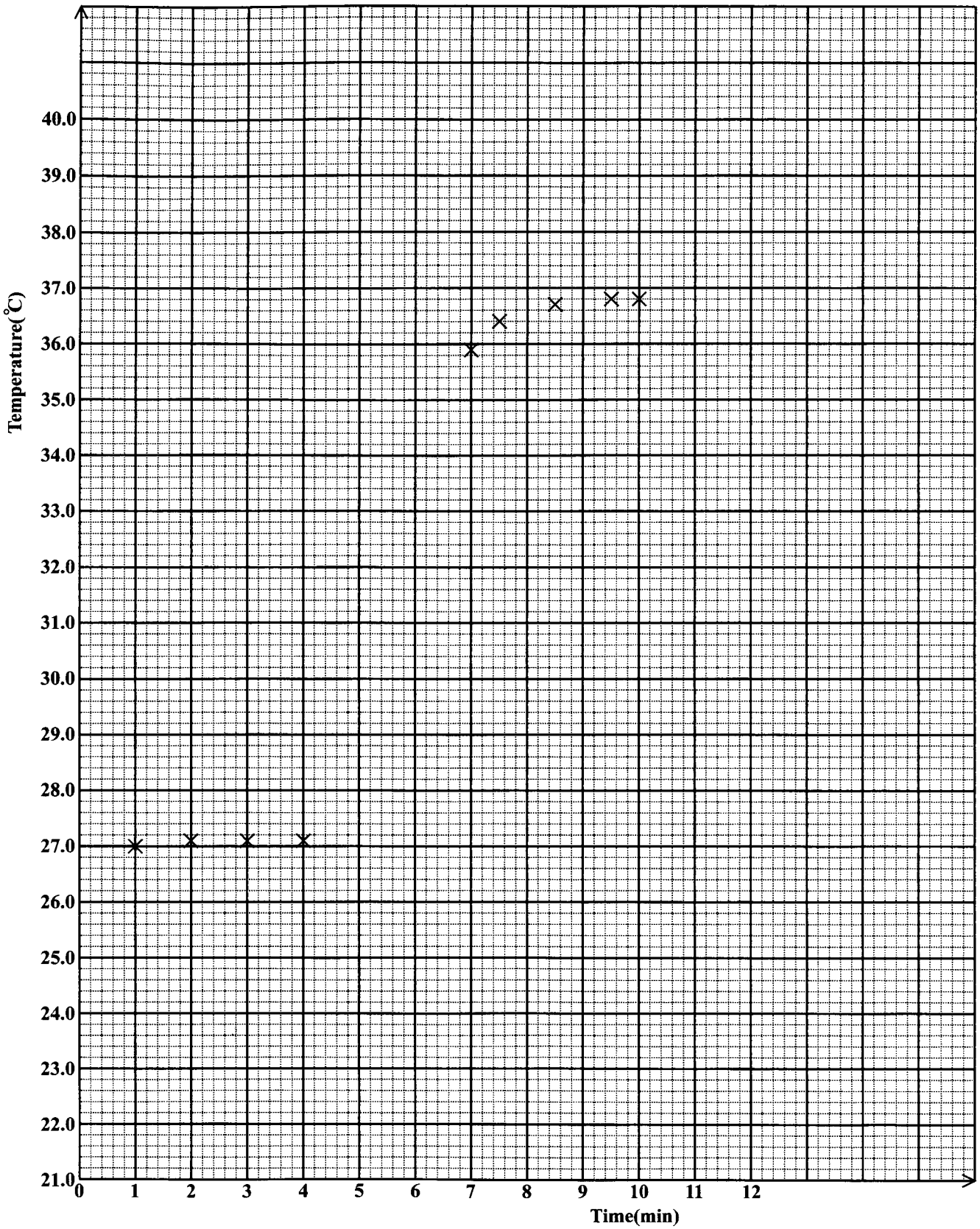

Figure 2. Graph of temperature against time

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(d) (i) The density of the final solution is 1.02 g cm^{-3}. Calculate the mass of the solution.

(1 mark)

(ii) The specific heat capacity of the final solution is 3.82 J g^{-1} $°C^{-1}$. Calculate the heat change for the reaction in the Styrofoam cup.

(2 marks)

(iii) Calculate the heat change for the formation of ONE mole of water.

(2 marks)

(e) (i) State ONE assumption that can be made when carrying out the experiment.

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(1 mark)

(ii) If the reaction was carried out in a glass beaker instead of a Styrofoam cup, how would this have affected the observed heat change?

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(2 marks)

(iii) Assuming that the reaction is carried out in a glass beaker, what precaution should be taken when stirring the reaction mixture?

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(iv) How should the procedure in (iii) above be modified?

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(2 marks)

Total 26 marks

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2. Bill conducted a number of tests on a solid mixture, W. The inferences made are recorded in Table 2. Complete Table 2 by inserting the observations based on the inferences made.

TABLE 2: TESTS ON MIXTURE W

	Test	Observation	Inferences
(a)	Distilled water was added to a portion of W and the resulting mixture stirred and filtered. The filtrate was divided into three parts and tests (b) to (d) done **on separate portions.**		
(b)	Aqueous sodium hydroxide was added dropwise and then in excess.	• • **(2 marks)**	Cu^{2+} ions are present.
(c)	Aqueous ammonium hydroxide was added dropwise and then in excess.	• • **(2 marks)**	Cu^{2+} ions are present.
(d)	Aqueous barium chloride was added, followed by dilute nitric acid.	• • **(2 marks)**	SO_3^{2-} ions are present.
(e)	Dilute nitric acid was added to the **residue** from (a), and any gas given off was tested.	• • **(2 marks)**	Hydrogen gas was evolved.
(f)	To the solution from (e), sodium hydroxide was added dropwise and then in excess.	• • **(2 marks)**	Zn^{2+} ions are present.

Total 10 marks

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3. A sample of sea water which was contaminated by petroleum was taken to the school chemistry laboratory for use in an experiment. Plan and design a suitable experiment to obtain pure water from the sample of sea water.

In your response you should include the following:

(a) Aim

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(1 mark)

(b) Materials

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(2 marks)

(c) Procedure

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(3 marks)

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(d) A fully labelled diagram of the apparatus required to obtain the pure water in this experiment

(4 marks)

(e) TWO possible sources of error OR precautions taken in your experiment

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(2 marks)

Total 12 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.